

## CLINTON, AR, USA

WMO: 720401

Lat: 35.600N

Lon: 92.450W

Elev: 516

StdP: 14.42

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00133

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>15.6 | (c)<br>19.1 | (d)<br>1.1                    | (e)<br>5.9 | (f)<br>22.0 | (g)<br>7.0 | (h)<br>8.0 | (i)<br>26.3 | (j)<br>18.0           | (k)<br>46.6 | (l)<br>16.3 | (m)<br>44.5 | (n)<br>1.7               | (o)<br>320 | (p)<br>0.373 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>21.9            | (c)<br>98.6     | (d)<br>76.7 | (e)<br>94.8 | (f)<br>77.1 | (g)<br>91.5 | (h)<br>76.7 | (i)<br>80.8         | (j)<br>90.2 | (k)<br>79.5 | (l)<br>89.5 | (m)<br>78.4 | (n)<br>88.3 | (o)<br>4.4           | (p)<br>240 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.9                            | 152.8 | 85.8 | 77.0 | 143.3 | 84.4 | 75.3 | 135.2 | 82.7 | 44.8          | 90.6 | 43.3 | 89.6 | 42.4 | 88.5 | 89.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c)  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| 14.7              | 12.0 | 10.4 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
|                   |      |      | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | 8.5                        | 83.4 | 6.0                | 2.3 | 4.2                                                | 85.0 | 0.7        | 86.4 | -2.7       | 87.6 | -7.1       | 89.3 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 59.9        | 36.8  | 42.7  | 50.7  | 59.5  | 68.1  | 77.4  | 79.6  | 79.0  | 72.6  | 60.7  | 49.5  | 41.2  |
|                                                                        | DBStd       | 16.72       | 9.38  | 10.03 | 8.99  | 7.27  | 6.94  | 4.47  | 4.62  | 4.85  | 6.05  | 8.57  | 8.36  | 8.67  |
|                                                                        | HDD50       | 1206        | 423   | 246   | 101   | 12    | 2     | 0     | 0     | 0     | 0     | 12    | 109   | 301   |
|                                                                        | HDD65       | 3595        | 874   | 624   | 447   | 189   | 53    | 0     | 0     | 0     | 9     | 188   | 470   | 741   |
|                                                                        | CDD50       | 4823        | 14    | 43    | 123   | 298   | 563   | 823   | 919   | 898   | 678   | 344   | 93    | 27    |
|                                                                        | CDD65       | 1737        | 0     | 0     | 4     | 26    | 149   | 373   | 454   | 433   | 237   | 55    | 4     | 2     |
|                                                                        | CDH74       | 17867       | 0     | 13    | 102   | 373   | 1396  | 3902  | 4853  | 4376  | 2202  | 581   | 40    | 29    |
|                                                                        | CDH80       | 8221        | 0     | 0     | 9     | 51    | 424   | 1875  | 2515  | 2222  | 953   | 148   | 4     | 20    |
| (14) Wind                                                              | WSAvg       | 2.3         | 2.7   | 3.1   | 3.4   | 3.5   | 2.5   | 2.1   | 1.5   | 1.4   | 1.4   | 1.9   | 2.2   | 2.5   |
| (15) Precipitation                                                     | PrecAvg     | 52.4        | 4.1   | 3.6   | 4.9   | 5.5   | 5.2   | 3.9   | 3.7   | 3.1   | 4.3   | 4.8   | 4.6   | 4.8   |
|                                                                        | PrecMax     | 74.5        | 7.4   | 7.9   | 12.6  | 11.6  | 11.1  | 6.5   | 7.3   | 7.9   | 9.6   | 14.9  | 12.0  | 7.4   |
|                                                                        | PrecMin     | 36.5        | 0.9   | 1.2   | 1.9   | 1.3   | 1.1   | 2.0   | 1.5   | 0.7   | 0.0   | 0.0   | 1.5   | 0.2   |
|                                                                        | PrecStd     | 10.4        | 1.8   | 1.9   | 2.7   | 2.3   | 2.6   | 1.3   | 1.5   | 1.8   | 2.4   | 3.6   | 2.7   | 2.1   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 68.4  | 76.6  | 81.2  | 84.1  | 90.3  | 101.5 | 102.0 | 101.6 | 96.6  | 88.1  | 79.2  | 72.4  |
|                                                                        |             | MCWB        | 61.1  | 63.8  | 64.7  | 70.1  | 72.5  | 72.7  | 74.8  | 77.9  | 76.2  | 72.1  | 65.9  | 64.9  |
|                                                                        | 2%          | DB          | 63.8  | 70.0  | 77.1  | 81.4  | 87.8  | 94.9  | 98.6  | 97.5  | 91.4  | 84.0  | 73.0  | 65.8  |
|                                                                        |             | MCWB        | 56.1  | 59.7  | 62.2  | 65.0  | 72.5  | 76.7  | 76.1  | 78.1  | 75.5  | 72.1  | 61.2  | 60.8  |
|                                                                        | 5%          | DB          | 58.5  | 65.6  | 72.5  | 78.9  | 84.2  | 91.4  | 94.6  | 93.4  | 89.9  | 81.1  | 69.8  | 61.1  |
|                                                                        |             | MCWB        | 50.3  | 57.7  | 59.9  | 64.0  | 72.0  | 76.3  | 77.8  | 76.2  | 75.6  | 67.6  | 61.3  | 53.9  |
|                                                                        | 10%         | DB          | 54.2  | 61.4  | 68.0  | 75.1  | 81.8  | 90.0  | 91.2  | 90.9  | 86.0  | 76.8  | 65.5  | 55.1  |
|                                                                        |             | MCWB        | 47.4  | 54.4  | 57.7  | 62.5  | 70.4  | 76.1  | 77.7  | 76.4  | 73.2  | 65.6  | 58.5  | 50.4  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 63.8  | 65.9  | 67.7  | 71.9  | 77.0  | 80.9  | 82.9  | 82.5  | 79.2  | 76.1  | 69.3  | 66.8  |
|                                                                        |             | MCDB        | 66.5  | 71.9  | 78.4  | 80.6  | 85.3  | 90.8  | 91.3  | 92.3  | 90.3  | 84.1  | 74.0  | 70.2  |
|                                                                        | 2%          | WB          | 59.4  | 63.1  | 65.4  | 69.2  | 75.2  | 79.5  | 81.0  | 80.8  | 77.9  | 74.0  | 65.9  | 62.2  |
|                                                                        |             | MCDB        | 62.8  | 66.5  | 74.4  | 77.0  | 83.1  | 89.4  | 89.5  | 90.7  | 88.1  | 81.5  | 70.6  | 64.8  |
|                                                                        | 5%          | WB          | 52.2  | 59.4  | 62.6  | 67.7  | 73.8  | 78.3  | 79.5  | 79.4  | 76.6  | 71.6  | 62.5  | 56.0  |
|                                                                        |             | MCDB        | 56.1  | 64.0  | 69.0  | 74.3  | 81.3  | 88.2  | 89.1  | 89.6  | 85.9  | 77.0  | 66.9  | 59.2  |
|                                                                        | 10%         | WB          | 46.5  | 54.8  | 59.7  | 65.8  | 72.3  | 77.3  | 78.3  | 78.1  | 75.1  | 68.5  | 59.1  | 52.3  |
|                                                                        |             | MCDB        | 52.0  | 59.5  | 65.6  | 71.6  | 79.5  | 87.0  | 88.0  | 88.0  | 83.2  | 73.4  | 64.5  | 54.9  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 21.7  | 21.9  | 24.2  | 25.3  | 22.0  | 22.1  | 21.9  | 21.9  | 23.0  | 25.5  | 24.1  | 19.6  |
|                                                                        |             | MCDBR       | 29.8  | 28.9  | 31.5  | 29.7  | 25.9  | 25.7  | 25.7  | 26.6  | 26.5  | 27.7  | 29.1  | 27.2  |
|                                                                        | 5% WB       | MCWBR       | 20.3  | 20.4  | 18.4  | 15.6  | 12.9  | 10.3  | 8.6   | 9.7   | 11.3  | 14.1  | 18.7  | 20.4  |
|                                                                        |             | MCWBR       | 22.8  | 23.4  | 26.7  | 22.3  | 20.9  | 21.9  | 21.3  | 22.7  | 22.9  | 21.8  | 24.1  | 23.0  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.317 | 0.322 | 0.357 | 0.391 | 0.426 | 0.451 | 0.483 | 0.477 | 0.405 | 0.356 | 0.328 | 0.307 |
|                                                                        |             | taud        | 2.484 | 2.467 | 2.403 | 2.309 | 2.270 | 2.214 | 2.176 | 2.168 | 2.357 | 2.475 | 2.504 | 2.526 |
|                                                                        | Ebn at Noon |             | 280   | 292   | 289   | 283   | 273   | 266   | 256   | 255   | 270   | 276   | 274   | 278   |
|                                                                        |             | Edh at Noon | 27    | 30    | 35    | 41    | 43    | 45    | 47    | 46    | 36    | 30    | 26    | 24    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 775   | 943   | 1294  | 1627  | 1774  | 2024  | 1947  | 1800  | 1539  | 1177  | 838   | 676   |
|                                                                        | RadStd      |             | 67    | 120   | 93    | 117   | 147   | 151   | 121   | 117   | 121   | 152   | 92    | 76    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (26 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page